

VSENS

Group Members:

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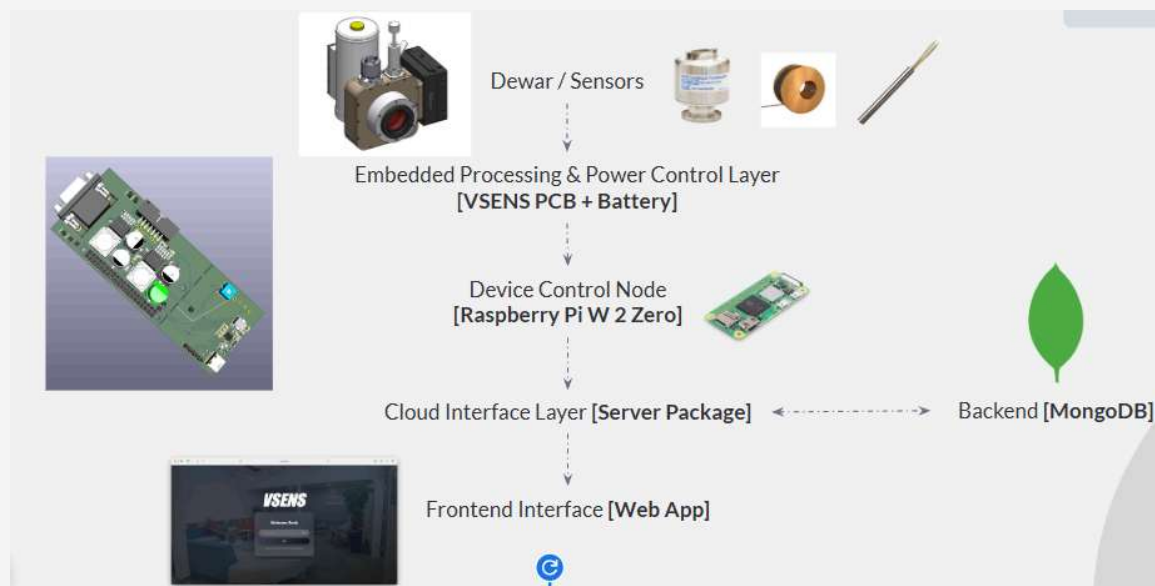
Final Progress

PROBLEM STATEMENT

Cryogenic Sensor Test Units (STUs) are vital in the development and characterization of infrared imaging devices. These STUs rely heavily on high-quality vacuum conditions to operate correctly, and monitoring these conditions is essential.

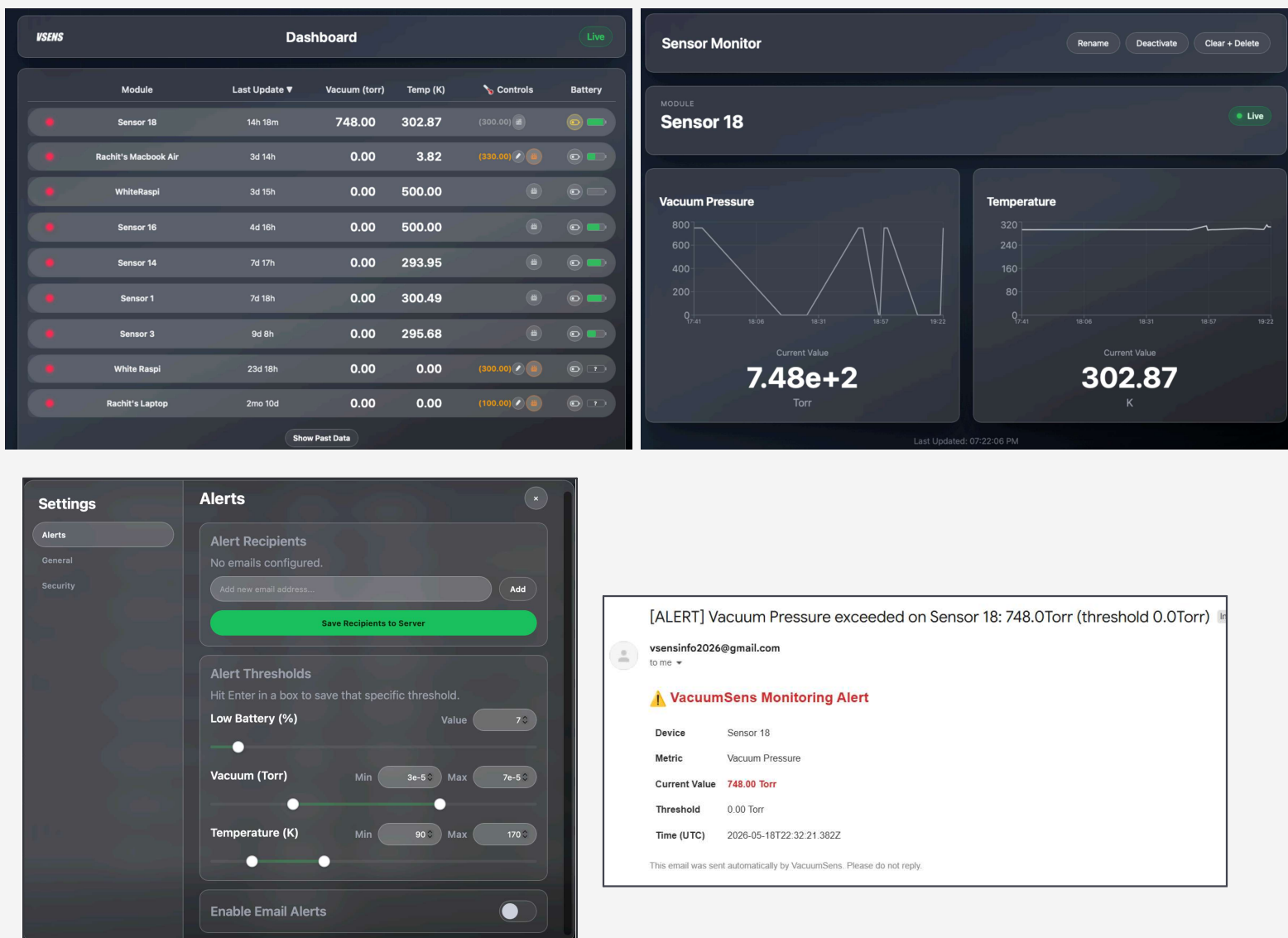
OUR GOAL

In this project, we will develop a networked system to remotely monitor the vacuum level across multiple STUs in the lab environment. Our system will continuously record and upload the sensor data and will raise an alarm if the vacuum level drops below certain thresholds. Different from traditional sensor setups, the idea is to have a centralized monitoring system that can take data from multiple sensors in a local area and compile them into a database for analysis or error detection. Furthermore, we will implement a temperature controller to maintain a slightly higher temperature than what the cryogen will provide. Finally, we will power everything with a rechargeable battery that allows us to run our system without being plugged into the wall. Everything should be easily accessible in a centralized user interface.



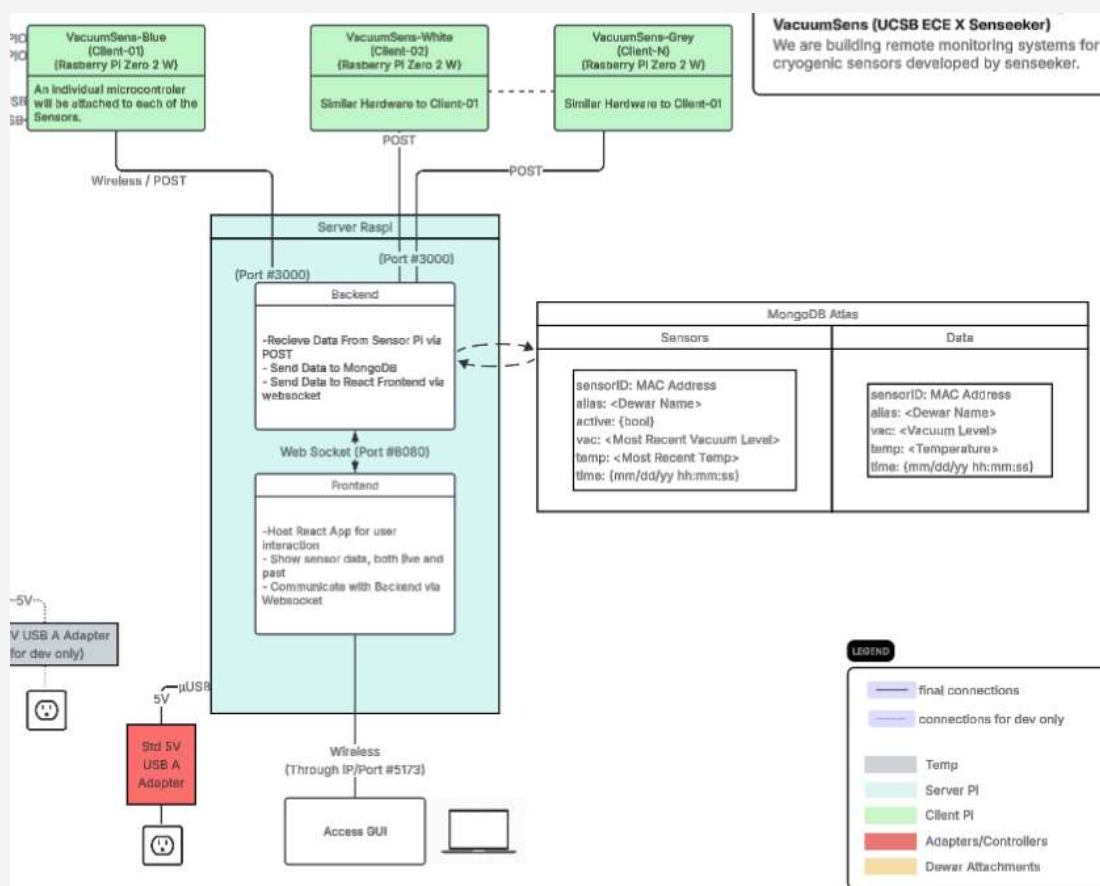
USER INTERFACE

The VSENS user interface is designed as a centralized dashboard for monitoring multiple sensor units in real time, prioritizing clarity, speed, and ease of use. It provides a unified view of each unit's status, including vacuum, temperature, battery level, and last update time, along with live/offline indicators to quickly identify issues. Users can access detailed views with real-time plots to observe trends and system behavior, while secure, password-protected controls allow safe management of nodes such as renaming or deactivating modules. Overall, the interface enables efficient, reliable monitoring of distributed systems in a lab environment.



BACKEND DATA ARCHITECTURE

The backend architecture consists of a [Node.js](#) server that connects to each of the clients via HTTPS POST. This receives and processes incoming data from each of the sensors before sending it out to a non-SQL database (MongoDB) for data persistence. It also pushes data to the frontend immediately via WebSocket to allow for real-time updates. It has the logic to handle any renaming or deletion of the Sensor modules and also processes all of the callbacks from the frontend to serve data queries and requests to and from the database.



ENCLOSURE DESIGN

The enclosure has gone through multiple revisions alongside our changing hardware needs. Initially, it was designed to host the sensor, the Raspberry Pi, the commercial RS232 converter, and our PCB v1 for battery control. However, after discussing with Senseseeker about their ideal product, it has gone through a complete overhaul and has been redesigned to be a standalone adapter that plugs onto the sensor via the DSUB connector. As such, it now has a much smaller form factor, being approximately 2.7" x 2.7" x 5", and hosts the PCB v4 and Raspberry Pi, as well as has a separate compartment for the battery to provide some thermal relief. It is separated into 3 pieces: the front portion, which has the d-sub connector on it, the back portion with the USB-C port, and a battery compartment. There is also an optional sensor guide mount that can go on the front portion



Figure showing the Enclosure containing PCB v4, Raspberry Pi, and Battery Compartment

PCB DESIGN

Senseseeker V1: For the first revision, we designed a power system around a single cell Lithium Ion battery using the MCP73871 charge management controller. This IC was chosen to allow continuous system power while simultaneously charging the battery via a USB C input. To meet our system voltage requirements, we utilized two MT3608 step up boost converters to generate a 5V rail for the Raspberry Pi and a 12V rail for our sensor. This board served as an initial proof of concept to validate our charging architecture before adding further functionality. Unfortunately, the board did not function as expected due to reverse placed Schottky diodes and wiring design errors for the MCP73871.

Senseseeker V2: Was designed to resolve the hardware bugs from V1 and upgrade out of spec components to better meet our design requirements. Realizing the 1.5A limit of the MCP73871 was insufficient to reliably run the system and charge the battery at the same time, we upgraded to the BQ25606RGER charging IC, which supports a current draw of up to 3A. We integrated female headers to mount the Raspberry Pi directly as a HAT, and expanded the board's capabilities by adding a 3.3V voltage regulator, a 16 bit ADC, a fuel gauge IC, an RS232 serial transceiver, and a current source to measure temperature via a diode. Despite these improvements, testing revealed issues with charging the battery.

Test PCB: To thoroughly debug our power architecture, we designed this Test PCB with a modular layout, physically separating our previous PCB so they can be tested and validated individually. In addition to our previous components, we introduced a new 3 cell Lithium Ion charging architecture paired with a USB C Power Delivery circuit capable of supplying up to 8A. We also included the original MCP73871 on this board with corrected schematic wiring to definitively test its functionality in isolation. We sent the gerber files to manufacture this validation board before Spring Break so that testing can inform us in the creation of our final design: Senseseeker V3.

Final PCB (V4): After all our iterations and breakout board testing, we finally landed on a complete design that included all the voltage regulators and charging circuit chips that worked. In the end, it was the most compact form factor to meet the sizing requirements, as well as a 3 cell design to fix our voltage requirements. Refer to the PCB section to see exactly what was included.

BILL OF MATERIALS

#	Component	Description	Qty	Total	Supplier
1	Custom PCB	Main system PCB	1	\$50.00 ¹	JLC PCB
2	Raspberry Pi Zero 2W	Main compute module	1	\$15.00	DigiKey
3	3D Printed Enclosure	PLA, custom design	1	\$1.75 ²	In-house
5	18650 Batteries	Supply a high voltage of 11.1V	3	\$15.00	Amazon
6	3D Printed Battery Holder	Will house all 3.7V batt. in series	1	\$1.00 ²	In-house
7	14-Pin IDC Ribbon Cable	14 Pin 2.54mm Pitch 30cm Len.	1	\$2.25	Amazon
TOTAL				\$85.00	
External Components					
8	MKS 901-P Vacuum Sensor	Vacuum Sensor	1	N/A	Senseeker
9	Vacuum Pump	Used For Development Only	1	N/A	Amazon
10	USB-C Power Adapter + Cable	System charging port	1	N/A	Amazon
11	DT-670 Silicon Diode	Temperature Sensor	1	N/A	Senseeker
12	HTR-25-100	Heater	1	N/A	Senseeker

Table 1 - BOM for one VSENS unit. Unit costs are approximate at the time of ordering.

¹ PCB fabrication costs assume a batch of 5 from JLCPCB. Per-unit cost decreases with higher volume.

² 3D printing costs are estimated at filament-only for in-house production.

VSENS Monitoring System: Core Features

Real-Time Environmental Monitoring

- Precision Sensing: Continuously tracks critical vacuum levels and temperature variations using highly reliable, integrated sensors.
- Cryogenic Compatibility: Specifically optimized to monitor operating conditions for cryogenic Sensor Test Units (STUs).

Automated Temperature Regulation

- Closed-Loop Control: Features an active temperature control loop utilizing a coordinated heater and temperature diode system.
- Dynamic Feedback: Maintains stable environmental parameters through real-time feedback, designed to simulate and operate reliably within Dewar environments.

Centralized Remote Dashboard

- Multi-STU Visualization: A unified frontend interface that allows users to remotely monitor real-time sensor data across multiple test units simultaneously.
- Intuitive Status Display: Provides a clear, user-friendly layout highlighting live environmental readings, overall system health, and active alerts.

Proactive Threshold Alerting

- Automated Notifications: Features an integrated email alerting module that instantly notifies users if environmental factors (such as vacuum pressure) drop below predefined safe thresholds.
- Rapid Response: Ensures abnormal conditions are flagged immediately, maximizing system safety and hardware reliability.

Custom Hardware Infrastructure

- Purpose-Built PCB Architecture: Runs on a custom-designed printed circuit board engineered for efficient signal routing, seamless sensor integration, and reliable communication with the central processing unit.
- Optimized Reliability: Hardware layout has been rigorously tested and iterated to ensure maximum compatibility and stability in demanding environments.

Spring Quarter Changes

On/Off Button

Integrated a dedicated physical button to provide users with direct, reliable control over the system's power state.

Compact Enclosure

Developed a custom-fitted chassis designed to protect the internal electronics that includes a battery module.

Temperature Control

PID works with liquid nitrogen, however it is not strong enough to heat quickly, so it needs more power. Since the battery is only 12 volts, future iterations should have a step up converter to 24 or higher volts.

Project Manual

Client Side

1. Flash the SD card of the client Raspberry Pi using the Official Raspberry Pi Imager
2. Connect to Raspi wifi and set up local wifi information.
3. Assemble the Client Module and install batteries.
4. Plug the D-SUB connector and ribbon cable for the temperature controller into the dewar.

Server Side

Install using the dedicated Windows Installer.

Assembly

1. Put Raspi on header
2. Put into cap and screw on DSUB header
3. Put 14 pin connector through the hole, and mounting holes
4. Put switch on other end of the cap and connect to PCB
5. Connect batteries in series and connect to PCB
6. Put on the back end of the cap

RISK ANALYSIS AND MITIGATION STRATEGY

During the development of the VSENS system, several potential technical risks were identified related to hardware reliability, system integration, and long-term operation.

Charging Circuit Thermal Issues

One of the primary issues encountered during testing was excessive heat generation from the battery charging integrated circuit on the custom PCB. During operation, the charging IC was observed to reach very high temperatures within approximately two minutes of usage. This behavior was present in both Version 1 and Version 2 of the PCB design and indicates a potential issue related to power dissipation, charging current configuration, or thermal management in the circuit design.

Mitigation Strategy

To isolate and diagnose the issue, the team transitioned from testing on the integrated PCB to using a development PCB platform. This approach allows each subsystem—including the charging circuit, power regulation, and battery interface—to be tested independently. By separating the components, we can more easily identify the root cause of the thermal behavior and validate safe operating conditions before integrating the circuit back into the final PCB design.

This debugging process is scheduled to be completed by the first week of the Spring Quarter. Once the root cause is confirmed, appropriate modifications will be applied to the PCB design, which may include adjusting the charging current configuration, improving power dissipation, or selecting a more suitable charging controller.

System Reliability Testing

In addition to addressing the charging circuit issue, the team plans to perform extended runtime testing once the hardware is fully assembled. Continuous operation tests will allow us to monitor thermal behavior, power consumption, and overall system stability over longer periods. These tests will help ensure that the system can reliably operate in laboratory environments where the devices may remain active for extended durations.

Through these mitigation strategies, the team aims to resolve hardware-level risks early in the development process and ensure that the final VSENS system operates safely and reliably.

TEAM MEMBER CONTRIBUTIONS

The VSENS project was developed through close collaboration among all team members. While each member took the lead on certain aspects of the system, the design, testing, and implementation of both hardware and software components were carried out collectively through regular discussion, debugging sessions, and integration work.

Rachit Gupta - Frontend Development

Led the development of the user interface and monitoring dashboard, including the design of the web-based interface for visualizing vacuum, temperature, and system status. Also contributed to system integration and testing across the full stack.

Andrew Yanez - PCB Design and Power Systems

Led the design and iteration of the VSENS custom PCB, including schematic design, layout, and power subsystem integration. Worked with the team on hardware debugging, component selection, and system-level testing.

Kauri Mai - System Software, Backend Integration and Enclosure Design

Led the development of the backend architecture responsible for data collection and communication between sensor nodes and the monitoring interface. Contributed to system integration, deployment, and testing of the full monitoring pipeline. Designed and modeled the Enclosure system to house the client devices.

Forrest Zhou - Sensor Interfacing and Alerts

Led the implementation of vacuum sensor communication and contributed to the alerting system for abnormal vacuum and temperature conditions. Assisted with hardware interfacing and system debugging.

Abdullah Ahmed - Hardware Design and System Architecture

Led the development of the temperature sensing and heater control subsystem and contributed to the overall system architecture. Worked closely with the team on hardware integration, control loop testing, and system validation.

Throughout the project, all team members participated in system integration, testing, debugging, and design discussions to ensure the final VSENS system operates reliably as a complete hardware–software platform.

APPENDIX

DESIGN FILES

Github: <https://github.com/vacuumsens>

Final deliverables: <https://github.com/vacuumsens/Final>

Server, client, PCB, and cad files are all included, including read.me's for instructions